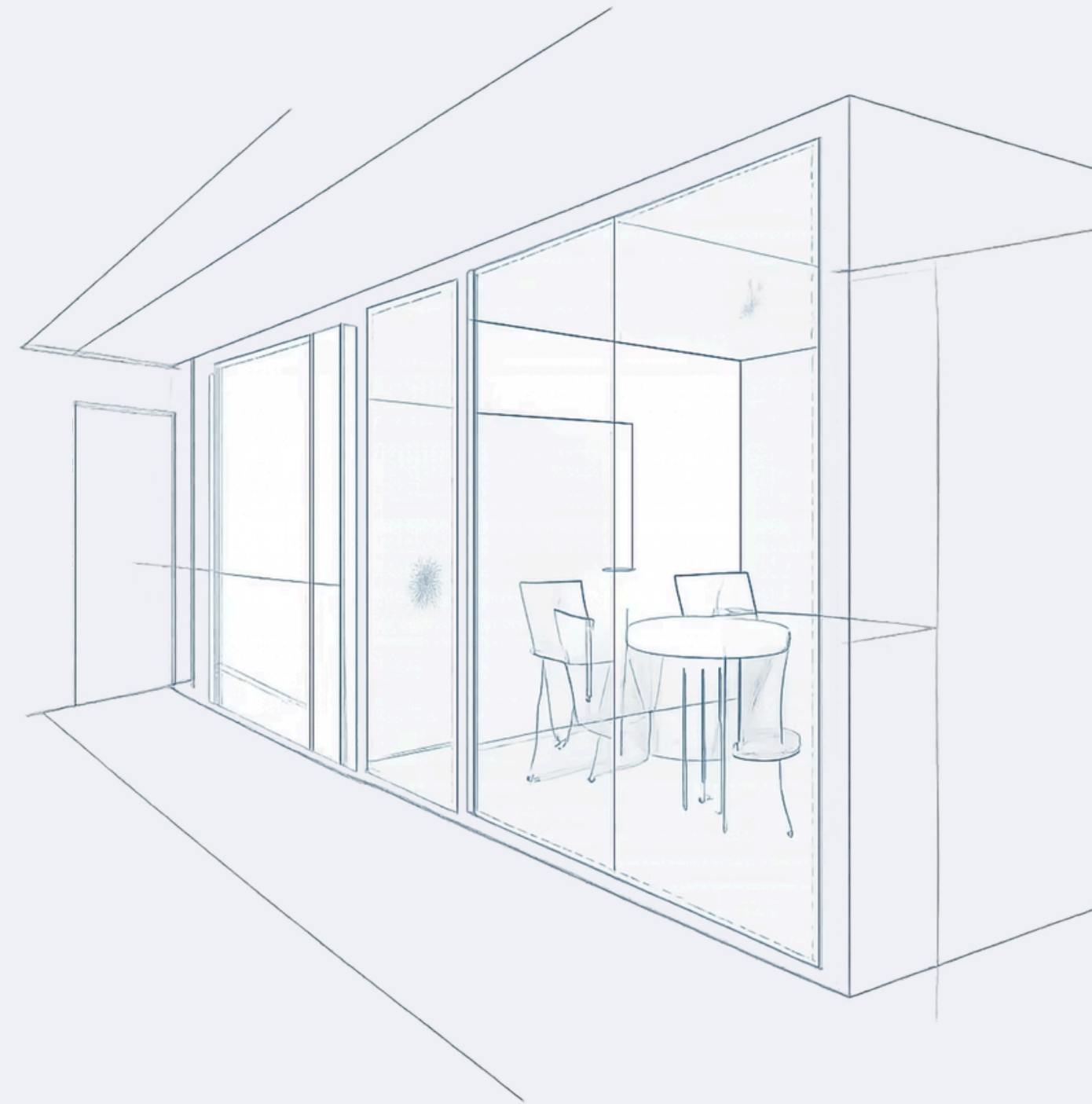


KOHLER Enterprise Intelligence Agent

Track 3: Unified Enterprise Conversational AI Agent Prototype —
KOHLER-MITWPU AI Research Lab case study. High-impact executive
summary of a permission-aware copilot for enterprise governance.

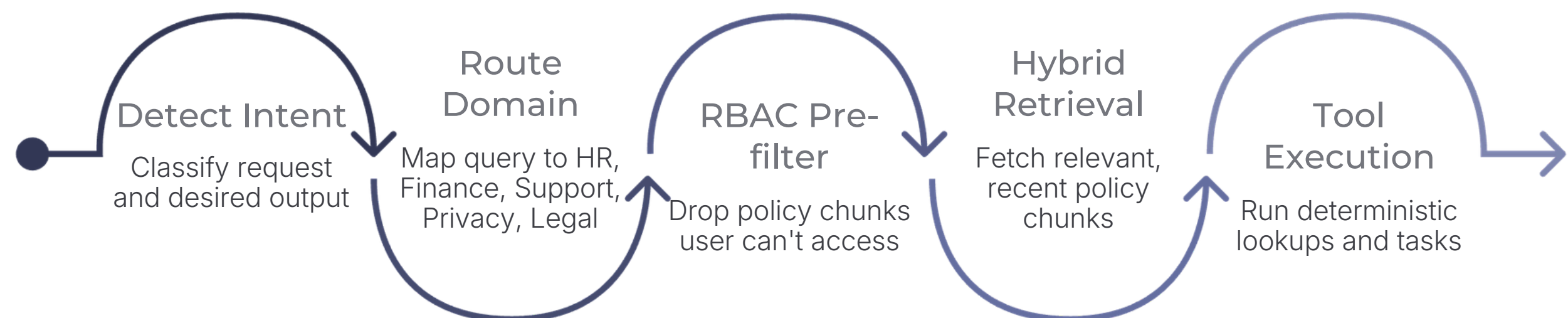


Core Approach

The KOHLER Enterprise Intelligence Agent runs every query through a 10-step governed loop, splitting work between deterministic code and the LLM:

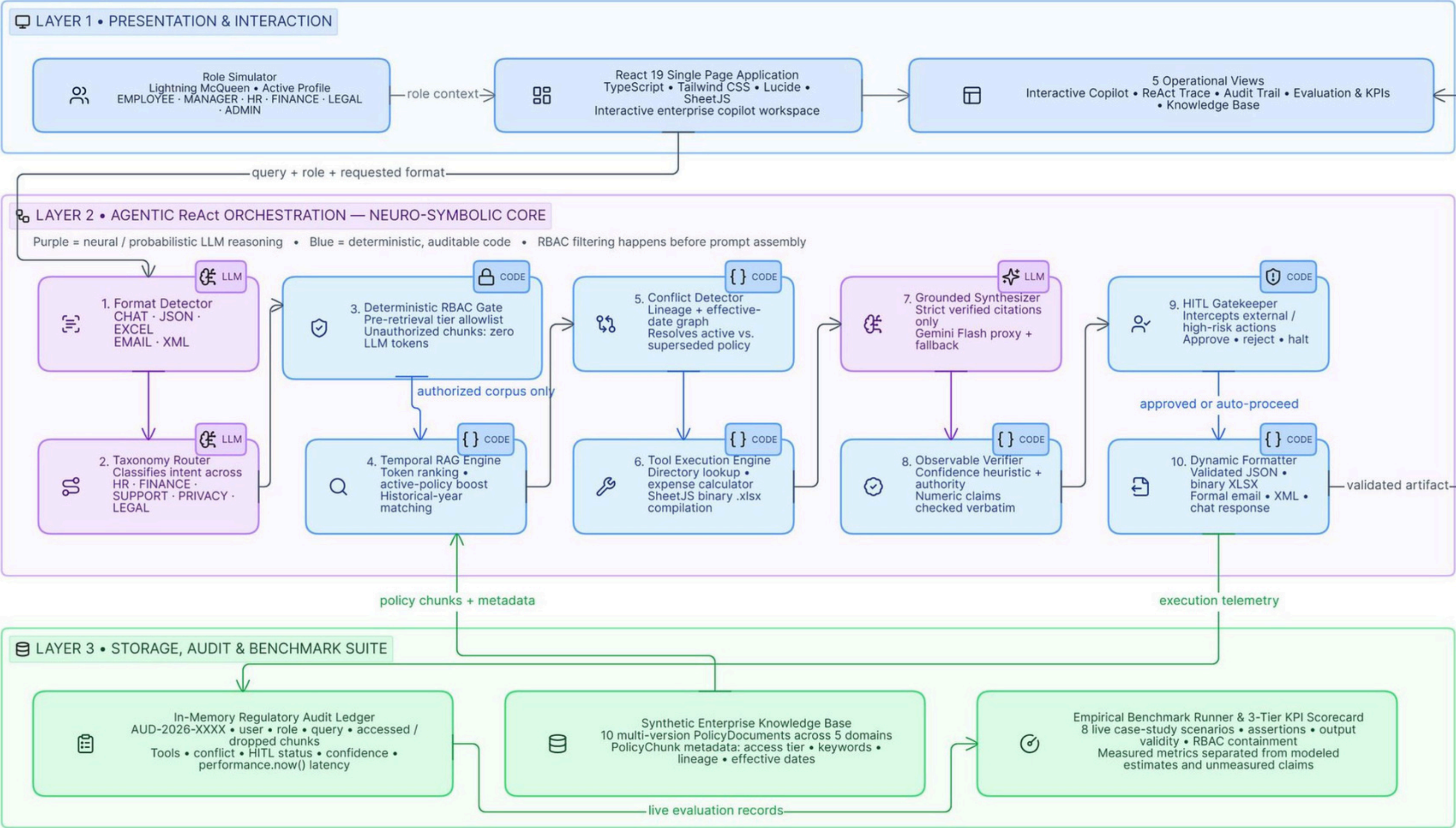
1. **Detect intent & format** – classify what's being asked and what output is needed (chat, JSON, Excel, email)
2. **Route domain** – map the query to HR / Finance / Support / Privacy / Legal
3. **RBAC pre-filter** (*deterministic*) – drop any policy chunk the user's role isn't cleared for, *before* it ever reaches a prompt
4. **Hybrid retrieval** – pull the most relevant, temporally-active policy chunks (token-overlap + recency scoring)
5. **Tool execution** (*deterministic*) – run any needed lookups (employee directory, expense audit, Excel builder)
6. **Conflict check** (*deterministic*) – detect if the query touches a superseded policy version and resolve to the active one
7. **Synthesize answer** – LLM writes the grounded response, citing only retrieved chunks
8. **Verify & score confidence** (*deterministic*) – check that every number in the answer actually exists in the source text; compute a weighted confidence score
9. **HITL gate** (*deterministic*) – if the action is high-risk (e.g. outbound email), pause for human approval
10. **Audit log** (*deterministic*) – record the query, decision, and latency

The one-line version: deterministic code guards security, dates, numbers, and approvals — the LLM only ever reasons over text it's already been cleared to see.



KOHLER Enterprise Intelligence Agent

Track 3 • Unified Enterprise Conversational AI Agent Prototype • KOHLER-MITWPU AI Research Lab Case Study



System Architecture

Six-tier access control, evaluated before retrieval

Role	Access Tier	Accessible	Restricted
EMPLOYEE	Standard Associate	Standard HR, travel, warranty terms	Executive severance, entertainment caps, legal logs
MANAGER	Dept. Manager	+ team approval thresholds, budgets	Executive compensation, CPO assessments
HR	HR Specialist	Full HR repository, severance formulas	Finance exception caps, legal records
FINANCE	Finance Auditor	Full travel/procurement, entertainment caps	HR health records, litigation files
LEGAL	Legal & Compliance	DPA's, whistleblower hotlines, risk	Unmasked compensation matrices
ADMIN	Enterprise Superuser	Full access — all 5 domains	None

FUNCTIONALITY

INNOVATION

The system is a **hybrid neuro-symbolic agent** (often called an **Agentic RAG system with a deterministic governance layer**).

Capability	Pure LLM / Naive RAG	Pure Expert System	This Hybrid Agent
Facts & grounding	Model weights — high hallucination risk	Hardcoded rules — brittle, rigid	RAG: retrieved from authoritative chunks
Access control (RBAC)	Prompt instructions only	Deterministic permission matrix	Deterministic — evaluated before retrieval
Supersession / conflicts	May blend 2024 and 2026 numbers	Explicit version graph	Deterministic metadata graph
Language & synthesis	Fluent, creative, flexible	Cannot handle unscripted language	LLM: cross-domain synthesis and drafts

Deterministic code guards the enterprise perimeter (security, dates, numbers, audit). The LLM provides fluent reasoning over grounded text. Neither side does the other's job.

TECHNICAL MOAT

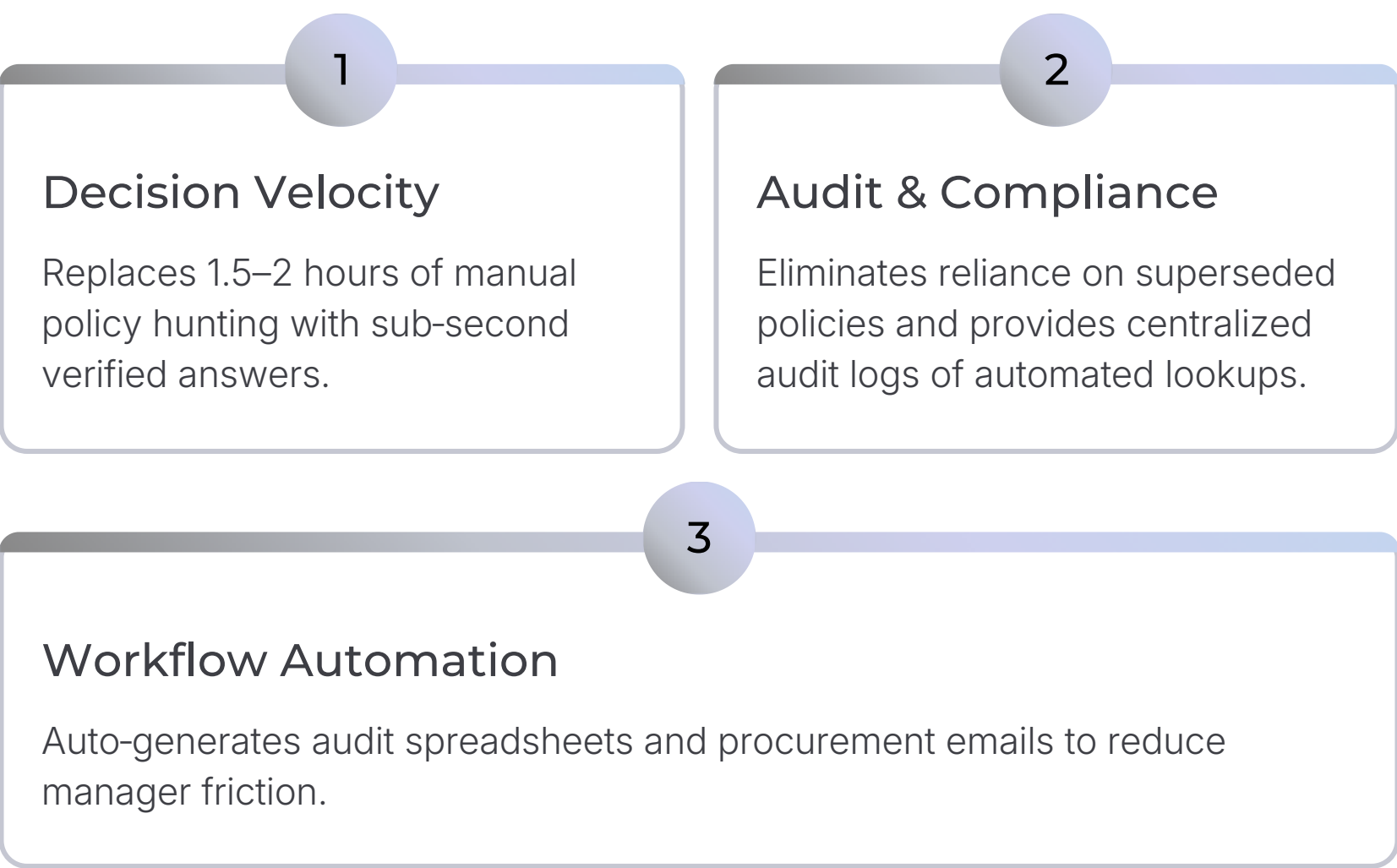
Rather than presenting an ungrounded LLM probability percentage, the system computes an observable composite score:

$$\text{Confidence} = 0.30 R + 0.25 A + 0.20 T + 0.15 C + 0.10 G$$

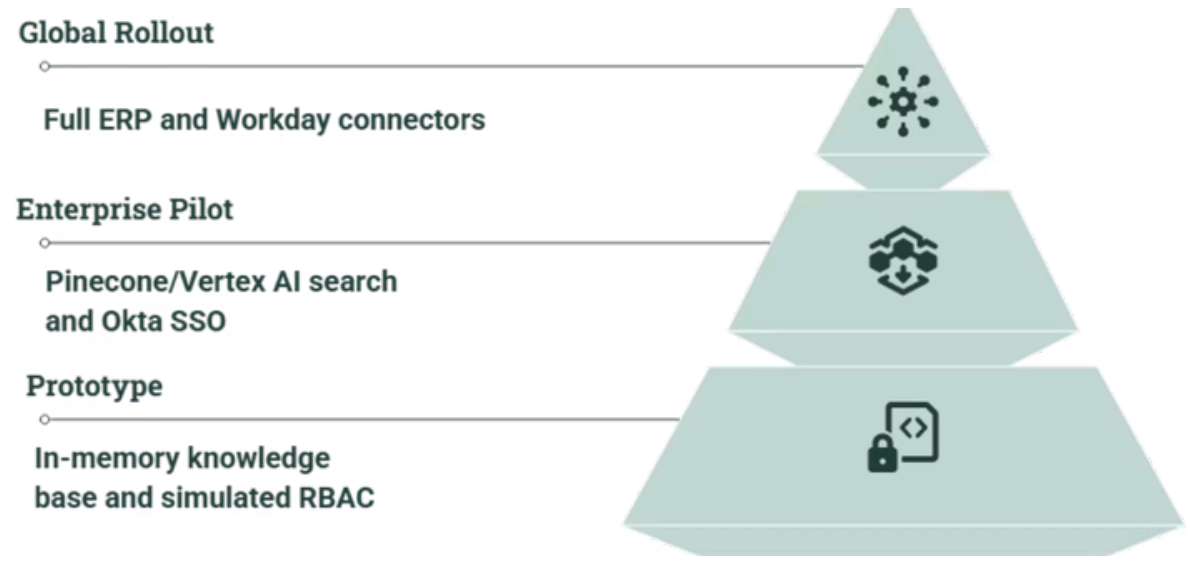
- R **(Retrieval Relevance)**: Percentage relevance of top retrieved chunks.
- A **(Authority Level)**: Document authority tier (Board = 100%, Corporate/Legal = 90%, Departmental = 75%).
- T **(Temporal Validity)**: Freshness relative to effective date and active status.
- C **(Conflict Freedom)**: 100% if no unresolved supersession conflicts exist, 70% if resolved.
- G **(Claim Grounding)**: Verification pass rate of extracted numerical clauses.

Users can click the Confidence badge in the UI to inspect the exact mathematical contributions of each factor.

Business Impact & Production Roadmap



The KOHLER agent pairs foundation models with deterministic security, temporal reasoning, and human-in-the-loop governance for a production-ready enterprise AI.



Takeaway: A mature vision for enterprise AI — deterministic access, temporal policy reasoning, and artifact synthesis enable secure, auditable, high-value automation.